bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

The propensity of fructose to induce metabolic dysfunction is dependent on the baseline diet, length of the dietary exposure, and sex of the mice.

Taghreed Fadhul[1], Se-Hyung Park[1,2], Heba Ali[2], Yasir Alsiraj[1,2], Jibran A. Wali[3,*], Stephen J. Simpson[3], and Samir Softic[1,2].

[1]Department of Pharmacology and Nutritional Sciences, University of Kentucky College of Medicine, Lexington, KY, 40536; tfa235@uky.edu. sh.park@uky.edu. yaal223@uky.edu. samir.softic@uky.edu.

[2]Department of Pediatrics and Gastroenterology, University of Kentucky College of Medicine, Lexington, KY, 40536; hhma223@uky.edu.

[3] Faculty of Science, School of Life and Environmental Sciences, The University of Sydney, Sydney, NSW, Australia; jibran.wali@rd.nestle.com. stephen.simpson@sydney.edu.au.

*Current address: Nestlé Institute of Health Sciences, Nestlé Research, Lausanne, Switzerland.

Corresponding author:

Samir Softic, MD

Associate Professor of Pediatrics

900 South Limestone, Wethington Rm 527, University of Kentucky

Lexington, KY 40536

Phone: (859) 218-1379; Fax: (859) 257-7799

E-mail: samir.softic@uky.edu

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

Abstract

Background/Objectives: Numerous studies have implicated high intake of sugar, particularly fructose, with the development of obesity and metabolic complications. On the other hand, fructose from fruits and vegetables has undisputed benefits for metabolic health. This paradox questions how the same fructose molecule can be associated with detrimental health effects in some studies and beneficial in others. **Methods:** To answer this question, male and female mice were fed different normal chow diets and provided 30% fructose solution in water. **Results**: Fructose-supplemented male mice on the Boston Chow Diet (BCD=23% protein, 22% fat, 55% carbs) gained weight, developed glucose intolerance and hepatic steatosis. In contrast, male mice on the Lexington Chow Diet (LXD=24% protein, 18% fat, 58% carbs) did not gain weight, remained glucose tolerant, and had normal hepatic lipid content when supplemented with fructose. Furthermore, fructose-fed male mice on a Low-Fat Diet (LFD=20% protein, 10% fat, 70% carbs) didn't gain weight, but once switched to the BCD, they gained weight, exhibited worsening liver steatosis, and more advanced hepatic insulin resistance. The effects of fructose are sex-dependent, as female mice didn't gain weight and remained insulin-sensitive when given fructose on BCD, despite developing hepatic steatosis. **Conclusions:** The differences in metabolic outcomes correlate with the propensity of the baseline diet to suppress hepatic ketohexokinase expression and the de novo lipogenesis pathway. This is likely driven by the dietary fat-to-carbohydrate ratio. Thus, metabolic dysfunction attributed to fructose intake is not a universal outcome; rather, it depends on the baseline diet, sex, and exposure length.

Keywords

Fructose, Ketohexokinase, De novo lipogenesis, Insulin resistance, Metabolic syndrome, MASLD, and Sex differences.

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

1. Introduction

There is a worldwide epidemic of obesity and metabolic dysfunction [1]. Increased intake of sugar and fat has been proposed as a key driver of these metabolic complications [2]. Indeed, numerous studies have documented the detrimental health effects of sugar intake [3-5], while reducing its consumption leads to improved health [6-9]. The majority of harmful effects of sugar have been attributed to its fructose component [10]. However, the detrimental metabolic outcomes of fructose intake are mainly observed in the context of co-ingestion with a high-fat diet (HFD). This brings into question whether fructose itself is sufficient to induce metabolic dysfunction or whether its effects are only observed in the setting of hypercaloric diets. This enigma is further exacerbated by the fact that fructose intake from fruits and vegetables appears to be protective from the development of metabolic dysfunction [11]. Thus, the question that remains to be answered is how the same molecule of fructose is associated with detrimental effects in one setting and beneficial effects in another.

One potential explanation acknowledges that fructose intake from solid food may be beneficial, while its consumption from sweetened drinks might be detrimental. When fructose is naturally found in solid foods, it comes together with fiber, vitamins, and antioxidants, which could lead to potential benefits by encouraging a feeling of fullness [12, 13], slowing down its absorption to enhance protective intestinal fructose metabolism [14-16] and neutralizing its metabolic sequela by scavenging oxidative stress [17]. Conversely, fructose from sugar-sweetened beverages (SSBs) is rapidly absorbed and lacks these protective elements, potentially causing excessive hepatic fructose metabolism to promote the development of hepatic steatosis and insulin resistance [5, 18, 19]. While highly intuitive, this explanation fails to account for studies showing that even liquid fructose intake leads to weight gain in some studies [20, 21], while in others, it does not support weight gain [22, 23]. More intriguing, some studies documented that fructose intake worsens glucose homeostasis [24], while others do not

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

substantiate this claim [25, 26] or even indicate that fructose consumption can improve serum glucose and insulin response [27]. These contradictory findings have led to controversy in the field of fructose metabolism and undermine the importance of dietary guidelines recommending a reduction in sugar intake [28, 29]. Our personal experience with this question came to the forefront when our chow-fed mice supplemented with 30% fructose solution failed to gain weight when our lab moved to Lexington, Kentucky, even though the same strain of mice in Boston, Massachusetts, consistently gained weight when provided fructose-sweetened water. These inconsistencies prompted the current study, which aims to explain how the background diets alter the metabolic consequences of fructose intake.

In this study, we set out to investigate how the intake of 30% fructose-sweetened water affects the development of obesity and metabolic complications in male and female mice fed different normal chow diets for different lengths of time. We found that the carbohydrate-to-fat ratio in the solid diet plays a significant role in how fructose affects the development of metabolic dysfunction. This is driven by the propensity of the baseline diet to suppress the expression of the ketohexokinase and enzymes regulating the hepatic de novo lipogenesis pathway. Moreover, the sex of the animal profoundly modifies these metabolic outcomes.

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

2. Materials and Methods

2.1. Animals and Diets

In accordance with NIH guidelines, all animal protocols were approved by the IACUC of the University of Kentucky, and Joslin Diabetes Center, where different experiments were performed. At Joslin Diabetes Center, mice were housed in cages with corncob bedding at 20-22 ºC on a 12 h light/dark cycle with *ad libitum* access to food and water. At the University of Kentucky, mice were housed in individually ventilated cages with woodchip bedding at 20-22°C on a 14 h light/10 h dark cycle with ad libitum access to food and water. At both institutions, C57BL/6J male mice, 6-8 weeks of age, were obtained from Jackson Laboratory and assigned to different diets. These diets included the Boston chow diet (Lab Diet, #5020), which contains 3.46 kcal/g metabolizable energy and consists of 23% protein, 22% fat, and 55% carbohydrates, and the Lexington chow diet (Teklad, #2018), which includes 3.1 kcal/g metabolizable energy and consists of 24% protein, 18% fat, and 58% carbohydrates. The mice were given a 30% fructose solution in water or regular water and remained on the respective diets for ten weeks. Additionally, at the University of Kentucky, we tested the effects of 30% fructose intake on a semi-purified low-fat diet (LFD) for 10 weeks, (Research Diets, #D12450K), which contains 3.85 kcal/g metabolizable energy consisting of 20% protein, 10% fat, and 70% carbohydrates, followed by a crossover to the BCD for an additional 10 weeks. The animals were weighed, and their food and water intake were recorded once per week. GTT and ITT were performed after 8 and 18 weeks on the diets. Mice were sacrificed between 8:00 and 11:00 am, with one mouse from each dietary group being euthanized before proceeding to the next mouse in the same cage. This process was repeated until all four mice per cage were sacrificed. The mice were injected with saline or 1U of insulin (Humulin R-Lilly #HI-213) via inferior vena cava 10 minutes before tissue collection in liquid nitrogen.

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

2.2. Glucose and insulin tolerance test

For the glucose tolerance test (GTT), mice were fasted overnight and injected intraperitoneally with 2g/kg of glucose. Blood glucose levels were measured at 0, 15, 30, 60, and 120 minutes using a glucose meter (Infinity, US Diagnostics). Insulin tolerance tests (ITT) were performed in nonfasted mice by i.p. injection of 1U/kg insulin per mouse. Blood glucose levels were measured at indicated times.

2.3. Liver Histology

Histology sections were prepared from formalin-fixed paraffin-embedded liver sections. H&E staining was performed by the University of Kentucky Pathology Research Core. Oil-Red O staining was performed to visualize hepatic neutral lipids. Frozen liver sections were air-dried for 30 minutes at room temperature, then fixed in 10% formalin for 15 minutes. After rinsing with PBS for 5 minutes, the slides were treated with 60% isopropyl alcohol for 5 minutes. Freshly prepared, filtered, Oil-Red O stain was applied for 10 minutes. Slides were rinsed with 60% isopropyl alcohol for 2 minutes, followed by three washes in distilled water. A coverslip was mounted using warmed glycerol gelatin.

2.4. Assays quantifying serum and liver metabolic dysfunction

Approximately 50 mg of liver tissue was weighed, with weights recorded since protein concentration measurement was not feasible with chloroform-methanol extracted samples. Samples were kept frozen on dry ice to prevent degradation. Each tissue sample was placed in a 2 mL tube with a steel bead and 300 µL of chloroform-methanol (2:1, v/v) and homogenized with a tissue lyser at 1/25 oscillations for 5 minutes. After adding 700 µL of chloroform (final volume 1 mL), the samples were homogenized again, then centrifuged at 15K for 15 minutes at 4°C. A 10 µL aliquot of the supernatant was transferred to glass culture tubes (12 x 75 mm) and evaporated in a fume hood at room temperature for 1–4 hours. Next, 200 µL of triglyceride

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

reagent (Pointe Scientific, #T7535-500) was added to each tube and briefly vortexed. Next, 200 µL of reagent was added to the wells for the standard curve, which was prepared using standards (0, 1, 2, 4, 6, 8, and 10 µL of TG solution at 200 mg/dL). All samples were incubated for 15 minutes at room temperature or 5 minutes at 37°C. Finally, 190 µL from each sample was transferred to a 96-well plate, and absorbance was read at 500 nm using a plate reader.

2.5. qPCR and mRNA quantification

mRNA was extracted by homogenizing liver tissue in TRIzol (Invitrogen, #15596018), treating it with chloroform, and participating in 70% ethanol. mRNA was purified using RNeasy Mini Kit columns (Qiagen, #74106). cDNA was made using a High Capacity cDNA Reverse Transcription Kit (Applied Biosystems, #4368813). qPCR was performed utilizing C1000 Thermal Cycler (BioRad, #CFX384) and QuantStudio™ 7 Flex Real-Time PCR System (TermoFisher Scientific, #4485701). Primers used for qPCR are listed in supplemental table 1.

2.6. Protein Extraction and Immunoblot

Tissues were homogenized in RIPA buffer (EMD Millipore with protease and phosphatase inhibitor cocktail (Bimake.com, #B14002, #B15002). Proteins were separated using SDS-PAGE and transferred to the Nitrocellulose membrane (BioRad, #1620112). Immunoblotting was achieved using the indicated antibodies listed in supplemental table 2. Images were captured by the ChemiDoc MP imaging system (BioRad, #12003154) and iBright imaging system (Thermo Fisher, #CL1000). Quantification of immunoblots was performed using ImageJ.

2.7. Plasma insulin levels and HOMA-IR quantification:

Following the manufacturer's protocol, plasma insulin levels were measured using an ultra-sensitive mouse insulin ELISA kit (Crystal Chem, #90080). Insulin resistance was assessed by calculating the Homeostasis Model Assessment of Insulin Resistance (HOMA-IR) index, using the formula: HOMA-IR = (Fasting Insulin [µU/mL] × Fasting Glucose [mg/dL]) / 405 × 10.

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

Specifically, the fasting glucose concentration was multiplied by the fasting insulin concentration, and the result was divided by 405, with the final value further multiplied by 10 to yield the HOMA-IR index.

2.8. Statistical Analyses

All statistical analyses and graph generation were conducted using GraphPad Prism version 10.3.0. Data are presented as mean ± SEM unless otherwise specified in the figure legends. The statistical tests for each data panel are detailed in the corresponding figure legends. Pairwise comparisons were conducted using Student's t-test, and analyses involving multiple groups were performed using ANOVA followed by Tukey's post hoc test. P-values <0.05 were considered statistically significant.

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

3. Results

3.1. The baseline diet modifies the impact of fructose on weight gain and insulin resistance.

To determine the impact of high fructose intake on different baseline diets, we placed C57BL6/J, male mice, on two different chow diets. The 9F 5020 chow diet (Boston chow diet) contains 3.46% calories per gram and consists of 23% protein, 22% fat, and 55% carbs. The T 2018 chow diet (Lexington chow diet) contains 3.1% calories per gram and consists of 24% of protein, 18% of fat, and 58% of carbs (Sup Figure 1A). The mice were given a 30% fructose solution in water or regular water and remained on the respective diets for ten weeks (Sup Figure 1B). Mice on the Boston chow diet supplemented with fructose (BCD+F) gained more weight (36.5±0.6g) than mice on the Boston chow diet drinking regular water (BCD+W) (28.8±1.4g) (Figure 1A). Interestingly, mice on the Lexington chow diet supplemented with fructose (LXD+F) did not gain additional weight (27.9±1.0g) compared to the mice on the Lexington chow diet drinking regular water (LXD+W) (27.5±0.9g) (Figure 1B). In agreement with weight gain, the BCD+F group consumed more calories (19.0±1.90kcal) than the mice on regular water (10.60±0.74kcal) (Figure 1C). However, the LXD+F group also consumed more calories (12.99±0.30kcal) than their water-supplemented counterparts (10.11±0.13kcal) (Figure 1D). Examining the sources of these calories, we found that mice given fructose water drank significantly more than those provided regular water (6.74±0.50ml vs. 3.80±0.22ml) (Sup Figure 1C). This increase in fructose intake did not affect solid food consumption in the BCD+F group compared to the BCD+W group (3.34±0.41g vs. 3.04±0.21g) (Sup Figure 1D). When the source of calories was further subdivided into protein, fat and carbohydrate, mice in the BCD+F group consumed more calories from carbohydrates, while caloric intake from protein and fat was unchanged (Sup Figure 1G). On the other hand, the mice on LXD also drank more fructose than regular water (4.84±0.13ml vs 3.32±0.14ml) (Sup Figure 1E), but, interestingly, solid food intake

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

was significantly reduced in the LXD+F-fed mice compared to the LXD+W-fed mice (2.50±0.05g vs 3.30±0.04g) (Sup Figure 1F). Correspondingly, mice in the LDX+F group consumed more calories from carbohydrates, but they showed a decrease in calories from protein and fat. A decrease in solid food intake in the LXD-, but not in the BCD-fed mice may account for the observation that these mice did not gain weight in response to fructose.

Next, we assessed glucose handling in these mice. Fasting insulin levels were not significantly altered in the BCD+F group (2.40±0.21ng/ml) compared to the BCD+W group (2.32±0.40ng/ml) (Figure 1E). Fasting blood glucose tended to be elevated in the BCD+F group (150.7±9.6ng/ml) compared to the BCD+W-fed mice (135.1± 9.9ng/ml) without reaching statistical significance (Figure 1F). The Homeostatic Model Assessment for Insulin Resistance (HOMA-IR) was not altered in the BCD+F-fed mice compared to the BCD+W-fed mice (8.90±0.84ng/ml vs 8.08±1.72ng/ml) (Figure 1G). On the other hand, the mice on the LXD+F diet (0.9±0.2ng/ml) had significantly lower serum insulin levels than those in the LXD+W group (1.8±0.3ng/ml) (Figure 1H). They also had reduced fasting blood glucose levels (118±5.4mg/dl vs 132.3±3.4mg/dl) (Figure 1I) and the LXD+F-fed group had lower HOMA-IR values (2.5±0.5ng/ml) compared to the LXD+W group (5.8±1.0ng/ml) (Figure 1J).

In line with these findings, the glucose tolerance test (GTT) area under the curve (AUC) indicated no significant differences between the BCD+F- and the BCD+W-fed mice (23655±1133mg/dl vs 25134±2171mg/dl) (Figure 1K, L). However, a trend for impaired insulin tolerance tests (ITT) (p=0.09) was observed in the area above the curve (AAC) for the BCD+F-group compared to the BCD+W-fed control (2348±586.5mg/dl vs 3534±300.0mg/dl) (Figure 1M, N). Moreover, blood glucose at 120 minutes of ITT was significantly elevated in BCD+F-fed mice compared to BCD+W-fed mice (Figure 1M). On the other hand, on LXD, the AUC for the GTT showed no significant differences between the LXD+F and LXD+W groups (33003±968.5ng/ml vs 31288±1054ng/ml) (Figure 1O, P). Similarly, the AAC from the ITT did

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

not demonstrate significant alterations between the LXD+F and LXD+W groups (4766±317.0ng/ml vs 4187±466.6ng/ml), although blood glucose was better in the LXD+F group at 15 minutes compared to the LXD+W group (Figure 1Q, R).

To evaluate hepatic insulin signaling, we injected 1 unit/mouse of insulin or saline into the inferior vena cava 10 minutes before sacrifice. The BCD+F group showed significantly decreased Akt phosphorylation compared to the BCD+W control group (Figure 1S, T). In contrast, insulin-stimulated Akt phosphorylation in the LXD+F group was not impaired compared to the LXD+W control group (Figure 1S, T). Furthermore, all mice on LXD had higher Akt phosphorylation than the mice on BCD. ERK phosphorylation increased across all mice treated with insulin, irrespective of the specific dietary composition. ERK phosphorylation was significantly lower in all mice fed LXD compared to the BCD (Fig. 1S, T), in agreement with insulin levels. In summary, fructose supplementation increased weight gain, impaired insulin tolerance at 120 minutes, and reduced hepatic insulin signaling on the BCD, while fructose intake on LXD did not induce weight gain and even improved some measures of insulin resistance.

3.2. Fructose supplementation on BCD, but not on LXD, supports hepatic steatosis.

To explain the differences in insulin resistance, we next investigated liver physiology. The liver weight of mice on the BCD supplemented with fructose was significantly higher compared to mice on regular water (1.60±0.06 vs 1.30±0.07) (Figure 2A). Conversely, the liver weight of mice on the LXD+F did not differ from mice on LXD+W (1.20±0.10 vs 1.20±0.05) (Figure 2B). In agreement with liver weight, BCD+F-fed mice developed hepatic steatosis compared to BCD+W-fed mice, as assessed by Oil Red O (Fig 2C) and Hematoxylin and Eosin (H&E) staining (Sup Figure 2A). Similarly, liver triglyceride (TG) accumulation was higher in the

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

BCD+F group compared to the BCD+W group (Figure 2D). In contrast, fructose supplementation in LXD-fed mice did not affect steatosis or TG accumulation compared to LXD+W-fed mice (Figure 2C, E).

Next, we investigated whether the difference in steatosis could be explained by the differences in hepatic fructose metabolism. The expression of Khk-c, the isoform of ketohexokinase that is primarily involved in the first step of fructolysis, was unchanged in BCD+F-fed mice compared to BCD+W-fed mice. Similarly, fructose supplementation in the BCD group did not alter the expression of Khk-a, the alternative isoform of ketohexokinase that plays a role in regulating fructose metabolism in different tissues (Figure 2F). However, there was a small but significant increase in Khk-c expression in the LXD+F-fed mice compared to the LXD+W-fed mice. Furthermore, there was a trend towards an increase in Khk-a expression in the LXD+F group, as determined by quantitative PCR (qPCR) (Figure 2G). We previously showed that fructose intake can alter protein levels via post-translational modifications, resulting in a poor correlation between protein and mRNA expression [30, 31]. Indeed, fructose supplementation in the BCD group resulted in elevated KHK-C protein levels compared to the BCD+W group (Figure 2H, Sup Figure 2B). Similarly, fructose supplementation in the LXD group led to an increase in KHK-C protein levels compared to the LXD+W group. Moreover, there was a significantly greater fold induction of KHK-C with fructose on BCD than on LXD when normalized to baseline expression in the water group for each diet (Sup Figure 2C). KHK-C protein in the liver is about 300 times more abundant than KHK-A [32, 33]. Accordingly, we only detected a faint KHK-A band, which was not affected by fructose supplementation of either BCD or LXD (Figure 2H, Sup Figure 2B, C). In summary, fructose intake in BCD more strongly induced KHK-C protein than in LXD, which was accompanied by greater steatosis in the BCD+F- compared to the LXD+F-fed mice.

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

Fructose consumption strongly stimulates hepatic de novo lipogenesis (DNL) [34]. In BCD-fed mice, fructose supplementation led to increased hepatic mRNA levels of enzymes involved in fatty acid synthesis, including ATP citrate lyase (Acly) and acetyl-CoA carboxylase 1 (Acc1), compared to BCD+W-fed mice (Figure 2I). Additionally, there was a trend toward increased fatty acid synthase (Fasn) expression, while the expression of stearoyl-CoA desaturase 1 (Scd1) remained unchanged. On the other hand, in LXD-fed mice, fructose supplementation elevated hepatic mRNA levels of Acly, Acc1, and Scd1 and tended to increase Fasn levels compared to LXD+W-fed mice (Figure 2J). The protein levels of ACLY, ACC1, FASN, and SCD1 were significantly increased in the BCD+F group compared to the BCD+W group (Figure 2K, Sup Figure 2D). Similarly, the protein levels of ACLY, ACC1, FASN, and SCD1 were significantly elevated in the LXD+F-fed mice compared to the LXD+W-fed mice. Consistent with a greater fat content of BCD than LXD, baseline protein levels of DNL enzymes were lower in the BCD compared to the LXD. This resulted in greater fold induction of the DNL proteins ACLY, ACC1, FASN, and SCD1 with fructose supplementation of the BCD than on the LXD (Sup Figure 2E), in agreement with greater KHK-C induction in this group.

Fructose intake can also reduce fatty acid β-oxidation (FAO). Fructose supplementation did not alter hepatic mRNA expression levels of key enzymes involved in FAO, including carnitine palmitoyl transferase 1A (Cpt1a), acyl-CoA oxidase 1 (Acox1), and acyl-CoA thioesterase 1 (Acot1) on either BCD (Sup Figure 3A) or LXD (Sup Figure 3B). However, the protein level of CPT1A, the rate-limiting enzyme of FAO, was significantly reduced in the BCD+F-fed mice compared to the BCD+W-fed mice (Figure 2L, Sup Figure 3C), whereas the protein levels of Acox1 and Acot1 remained unchanged. Likewise, in the LXD group, fructose supplementation led to a significant reduction in the protein level of CPT1A, with no significant changes observed in the levels of Acox1 and Acot1 when compared to LXD+W-fed mice (Figure 2L, Sup Figure 3C). Additionally, the levels of other enzymes involved in fatty acid β-oxidation,

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

including carnitine acylcarnitine translocase (CACT), organic cation/carnitine transporter 2 (OCTN2), carnitine palmitoyl transferase II (CPT2), acyl-CoA dehydrogenase long chain (ACADL), acyl-CoA dehydrogenase very long chain (ACADVL), and hydroxy acyl-CoA dehydrogenase trifunctional multienzyme complex subunit alpha (HADHA) were minimally affected with fructose supplementation in either BCD or LXD groups (Sup Figure 3D, E). In summary, fructose supplementation supports the development of hepatic steatosis only in mice on the BCD, but not in mice on the LXD. On a molecular level, fructose showed a stronger fold induction in the protein levels of KHK-C and enzymes mediating the DNL pathway on the BCD compared to the LXD.

3.3. Switching from a low-fat diet to a Boston chow diet restores weight gain and insulin resistance.

To test if the outcomes of fructose supplementation are solely dictated by the underlying baseline diet, we tested the effects of fructose intake on a semi-purified low-fat diet (LFD) for 10 weeks, followed by a crossover to the BCD for an additional 10 weeks (Sup Figure 4A). The LFD (D12492) provides 3.85 calories per gram, consisting of 20% protein, 10% fat, and 70% carbohydrates (Sup Figure 1A). Mice on the low-fat diet supplemented with fructose (LFD+F) exhibited a non-significant trend towards lower weight gain (30.0±0.6g) compared to those on the low-fat diet consuming regular water (LFD+W) (31.8±1.1g) (Figure 3A). Interestingly, upon switching to the BCD, fructose supplementation resulted in a significant increase in weight gain (43.1±1.0g) compared to BCD+W-fed mice (37.5±2.0g) (Figure 3B). Total caloric intake in the LFD+F group was not significantly changed (11.80±0.20kcal) than in the mice on regular water (11.70±0.22kcal) (Figure 3C), however, mice fed the BCD+F diet had a significant increase in caloric intake (14.01±0.44kcal) compared to the BCD+W group (12.20±0.13kcal) (Figure 3D). Moreover, mice given fructose on LFD drank significantly more than those provided regular

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

water (4.34±0.20ml vs 2.93±0.05ml) (Sup Figure 4B). In contrast, solid food consumption was significantly reduced in the LFD+F group compared to the LFD+W group (1.82±0.07g vs 3.03±0.05g) (Sup Figure 4C). The mice on BCD also drank more fructose than regular water (4.30±0.11ml vs 3.30±0.07ml) (Sup Figure 4D). Similar to the LFD groups, solid food intake was significantly decreased in the BCD+F-fed mice compared to the BCD+W-fed mice (2.60±0.04g vs 3.52±0.03g) (Sup Figure 4E). However, the decrease in solid food intake was greater on LFD (41%) than on BCD (28%). This indicates that fructose-supplemented mice experienced a relative increase in solid food intake when switched from LFD to BCD, which in part explains higher weight gain.

Fasting insulin level was markedly elevated in the LFD+F group compared to the LFD+W group (0.52±0.10ng/ml vs 0.22±0.04ng/ml) (Figure 3E). Moreover, the LFD+F group showed an increase in fasting blood glucose compared to the LFD+W-fed mice (164.1±6.94 mg/dl vs 140.3±6.90mg/dl) (Figure 3F), and HOMA-IR revealed a trend towards increased insulin resistance (2.13±0.60ng/ml vs 0.80±0.15ng/ml) (Figure 3G). Similarly, serum insulin levels in the BCD+F group (1.80±0.23 ng/ml) were significantly higher than those in the BCD+W group (0.90±0.35ng/ml) (Figure 3H). However, the BCD+F diet did not significantly alter fasting blood glucose levels (161.0±9.44 mg/dl vs 151.9±7.30mg/dl) (Figure 3I), and HOMA-IR values remained unchanged between the BCD+F-fed mice (7.01±1.20ng/ml) and the BCD+W-fed mice (4.11±1.40ng/ml) (Figure 3J).

The IPGTT revealed significantly higher AUC in the LFD+F group compared to the LFD+W group (34359±1268ng/ml vs 27788±1064mg/dl) (Figure 3K, L), and glucose levels during GTT were significantly higher in the LFD+F group at 15, 30, and 60 minutes compared to the LFD+W group (Figure 3K). Conversely, the ITT demonstrated no significant difference in the AAC between the LFD+F and LFD+W groups (3084±370ng/ml vs 2253±312mg/dl) (Figure 3M, N). On the other hand, the GTT AUC indicated a trend towards impaired glucose tolerance in

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

BCD+F-fed mice compared to the BCD+W-fed mice (51875±3072ng/ml vs 42713±3289mg/dl; p=0.06) (Figure 3O, P) and glucose levels were significantly higher in the BCD+F group at 60 and 120 minutes compared to the BCD+W group (Figure 3O). The overall ITT AAC showed no significant difference (5567±443mg/dl vs. 6573±430 mg/dl), but glucose levels were higher in the BCD+F-fed mice compared to the BCD+W-fed controls at 60 and 90 minutes (Figure 3Q, R).

Assessment of hepatic insulin signaling revealed fructose supplementation in the LFD group significantly reduced Akt phosphorylation compared to the LFD+W control group (Figure 3S, T). Similarly, fructose supplementation of the BCD-fed mice resulted in a significant reduction in insulin-stimulated Akt phosphorylation compared to the BCD+W-fed control. Overall, Akt phosphorylation was higher in all LFD-fed mice compared to those on BCD. Furthermore, ERK phosphorylation increased in all mice treated with insulin, regardless of dietary composition (Figure 3S, T). In summary, fructose supplementation led to increased weight gain when mice were transitioned from the LFD to the BCD. Interestingly, in both the LFD and BCD groups, fructose consumption impaired some aspects of glucose tolerance and insulin resistance compared to their respective controls. Moreover, in LFD-fed mice, adverse changes in glycemic status with fructose ingestion were observed despite no significant increase in body weight.

3.4. Fructose enhances hepatic steatosis more strongly on BCD than on LFD.

Liver weight was minimally increased in mice consuming a LFD supplemented with fructose compared to those provided with regular water (1.40±0.04g vs 1.09±0.09g) (Figure 4A). However, mice on the BCD supplemented with fructose demonstrated a greater increase in liver weight relative to those on regular water (2.01±0.20g vs 1.60±0.10g) (Figure 4B). Despite a

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

slight increase in liver weight, the LFD+F group did not exhibit an increase in hepatic steatosis compared to the LFD+W group, as evidenced by ORO staining (Figure 4C) and H&E staining (Sup Figure 5A). Consistent with higher liver weight, the BCD+F group developed hepatic steatosis compared to the BCD+W group (Figure 4C; Sup Figure 5A). In agreement with steatosis, hepatic TG accumulation was not affected in the LFD+F group compared to the LFD+W control (Figure 4D). In contrast, liver TG levels were elevated in the BCD+F group relative to the BCD+W control (Figure 4E).

Next, we assessed the effects of fructose on KHK levels on these diets. The expression of Khk-c and Khk-a remained unchanged in LFD+F-fed mice relative to LFD+W-fed controls. (Figure 4F). In contrast, Khk-c and Khk-a expression was upregulated in the BCD group following fructose supplementation compared to the BCD+W group (Figure 4G). At the protein level, fructose supplementation in the LFD group resulted in elevated protein levels of KHK-C compared to the LFD+W group (Figure 4H; Sup Figure 5B). Similarly, KHK-C protein levels were significantly elevated in BCD+F-fed mice relative to BCD+W-fed counterparts. Calculating fold-change over its baseline, fructose resulted in greater induction of KHK-C on BCD compared to LFD (Sup Figure 5C). Conversely, fructose supplementation did not induce a significant change in KHK-A protein levels in either the LFD or BCD group (Figure 4H; Sup Figure 5B, C).

In LFD-fed mice, fructose supplementation did not elicit significant changes in the hepatic mRNA levels of key enzymes involved in fatty acid synthesis, including Acly, Acc1, Fasn, and Scd1, when compared to LFD+W-fed controls (Figure 4I). In contrast, fructose supplementation in BCD-fed mice led to a significant upregulation of hepatic mRNA levels for Acly, Acc1, Fasn, and Scd1 relative to BCD+W-fed mice (Figure 4J). At the protein level, the LFD+F group exhibited increased levels of ACLY, ACC1, and FASN compared to the LFD+W group (Figure 4K; Sup Figure 5B). Similarly, in the BCD group, fructose supplementation resulted in a significant elevation of protein levels for ACLY, ACC1, FASN, and SCD1 compared

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

to the BCD+W-fed control (Figure 4K; Sup Figure 5B). Moreover, the BCD had lower baseline levels of DNL enzymes, so fructose supplementation resulted in a greater fold-increase in DNL enzymes on BCD than on LFD (Sup Figure 5C). This is consistent with the greater induction of KHK-C protein on the BCD than on the LFD.

In both LFD- and BCD-fed mice, fructose supplementation did not markedly affect hepatic mRNA levels of enzymes involved in fatty acid beta-oxidation, including Cpt1a, Acox1, and Acot1, when compared to regular water controls (Sup Figure 5D). At the protein level, CPT1A was not affected in LFD+F-fed mice compared to their LFD+W-fed counterparts (Figure 4L; Sup Figure 5F). Conversely, CPT1A was significantly decreased with fructose supplementation in the BCD group compared to the BCD+W group. In contrast, ACOX1 and ACOT1 protein levels remained unchanged with fructose supplementation on either diet (Figure 4L; Sup Figure 5F). In summary, fructose supplementation on BCD compared to the LFD enabled greater induction in KHK-C and DNL enzymes while lowering CPT1A protein, which may explain the development of hepatic steatosis in these mice.

3.5. Fructose supplementation results in weight gain and insulin resistance in male but not female mice.

In the subsequent experiments, we compared the effects of fructose supplementation in male and female mice. Like the previous experiment, the mice were on LFD for ten weeks and then crossed over to BCD for another ten weeks (Sup Figure 6A). As previously shown, male mice gained weight with fructose supplementation when crossed to BCD. In contrast, female mice did not gain weight with fructose supplementation on LFD (19.7±0.6g vs 20.9±0.6g), and similarly, their weight gain remained unchanged when switched to BCD compared to female mice receiving regular water (23.6±0.8g vs 25.5±1.4g) (Figure 5A). In agreement with body

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

weight, the mass of perigonadal (PG) adipose tissue increased in male mice supplemented with fructose, while it was unchanged in female mice (Sup Figure 6B). As expected, both male (4.30±0.11ml vs 3.30 ±0.07ml) and female (3.44±0.16ml vs 2.92±0.02ml) mice drank more fructose than regular water (Figure 5B). This was associated with a corresponding reduction in solid food intake in both male (2.60±0.04g vs 3.52±0.03g) and female mice (2.18±0.11g vs 4.30±0.10g) after 20 weeks on the diets (Figure 5C). However, female mice exhibited a greater reduction in solid food intake with fructose than males (50% vs 28%). While caloric intake from fat and protein was lower in both male and female mice supplemented with fructose, total carbohydrate intake was higher in males given fructose, but it was actually unchanged in female mice supplemented with fructose (Sup Figure 6C). Consequently, the total caloric intake significantly increased in male mice consuming fructose (14.01±0.44kcal vs 12.20±0.13kcal), while it decreased in female mice (11.40±0.30kcal vs 14.71±0.40kcal) given fructose relative to the regular water group (Figure 5D).

Unlike males, fasting insulin levels were not increased with fructose feeding in female mice (0.13±0.03ng/ml vs 0.11±0.02ng/ml), and insulin levels were much lower in females compared to male mice (Figure 5E). Blood glucose levels (Figure 5F) and HOMA-IR (Figure 5G) were not significantly affected in male or female mice exposed to fructose for 20 weeks. However, female mice had significantly lower blood glucose and HOMA-IR values than males. Similarly, female mice supplemented with fructose showed no significant change in glucose tolerance AUC (29449±1917ng/ml vs 32858±3285mg/dl) compared to those given regular water (Figure 5H, I), even though males showed a tendency to have impaired glucose tolerance with fructose intake. Female mice also demonstrated no significant difference in ITT AAC between the fructose and regular water groups (8106±258ng/ml vs 6472±1001mg/dl) even though blood glucose values tended to be lower at 30, 60, and 90 minutes in female mice fed fructose (Figure

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

5J, K). On the other hand, glucose values at 60 and 90 min were actually higher in male mice exposed to fructose.

When assessing hepatic insulin signaling, insulin-stimulated AKT phosphorylation remained unaltered in female mice supplemented with fructose compared to the regular water group (Figure 5L, M), despite a significant reduction in male mice. However, like the males, female mice given fructose exhibited a slight decrease in ERK phosphorylation compared to their water controls. In summary, fructose-fed female mice displayed lower weight gain and were protected from glucose intolerance and insulin resistance compared to male mice.

3.6. Fructose supplementation in female mice more strongly upregulates KHK-C and DNL enzymes than in males.

Similar to males, female mice supplemented with fructose showed a trend toward an increase in liver weight compared to those receiving regular water (1.40±0.09g vs 1.20±0.08g) (Figure 6A). When corrected for body weight, female mice fed fructose demonstrated a significant increase in the liver-to-body weight ratio (5.50±0.20g vs 4.53±0.20g) compared to female mice on regular water (Figure 6B). Consistent with liver weight, fructose supplementation in male-mice led to the development of hepatic steatosis. Similarly, female mice given fructose also developed hepatic steatosis, as indicated by ORO staining (Figure 6C) and H&E staining (Sup Figure 7A). Liver TGs were significantly increased in female mice given fructose compared to water controls (47.21±2.20g vs 19.09±2.30g), similar to an increase observed in fructose-fed male mice (Figure 6D). On the other hand, hepatic cholesterol levels were profoundly increased in female mice given fructose compared to water controls (3.70±0.20g vs 2.70±0.11g), while fructose-fed male mice only showed a slight and non-significant trend in hepatic cholesterol (2.80±0.22g vs 2.31±0.11g) (Figure 6E). In summary, female mice supplemented with fructose

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

develop a much larger increase in liver-to-body weight ratio, hepatic steatosis, and hepatic cholesterol levels than males.

Similar to males, KHK-C protein levels increased in female mice supplemented with fructose compared to their counterparts in regular water (Figure 6F, H; Sup Figure 7B). When corrected to their respective baseline levels in water groups, a fold-change increase in KHK-C with fructose intake was stronger in female than in male mice (Figure 6H). In contrast, the protein level of KHK-A did not change in male or female mice supplemented with fructose (Figure 6F, H; Sup Figure 7B). Furthermore, the protein levels of ACLY, ACC1, FASN, and SCD1 increased with fructose supplementation of both male and female mice (Figure 6G). Again, fold-change increase with fructose supplementation was stronger for ACLY and ACC1 in females compared to male mice (Figure 6G, H; Sup Figure 7B).

Similar to male mice, female mice supplemented with fructose exhibited a significant reduction in CPT1A protein compared to those receiving regular water (Figure 6I, J). Conversely, Acox1 and Acot1 protein levels showed no significant alterations in either male or female mice following fructose supplementation (Figure 6I, J). In summary, fructose supplementation in both male and female mice resulted in a significant increase in KHK-C protein levels, along with elevated levels of ACLY, ACC1, FASN, and SCD1 proteins. The increase in KHK-C and DNL enzymes was stronger in female mice, in agreement with a higher liver-to-body weight ratio and liver steatosis.

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

4. Discussion

Our study examined how fructose intake, in the context of different baseline diets, varying durations of dietary exposure, and sex differences, impacts metabolic health and insulin resistance. We found that male mice drinking 30% fructose solution on the Boston chow diet for ten weeks gain weight, develop hepatic steatosis, and have impaired hepatic insulin signaling. Remarkably, the mice consuming the same concentration of fructose on the Lexington chow diet do not gain additional weight, have normal lipid content in the liver, and remain insulin sensitive. To investigate whether altering the baseline diet is sufficient to drive these differences, we placed fructose-supplemented mice on a low-fat diet for ten weeks, then crossed them over to BCD for another ten weeks. The mice on LFD do not gain weight or develop hepatic steatosis when exposed to fructose, but they exhibit hepatic insulin resistance. Indeed, once switched to the BCD, the mice gain significantly more weight, exhibit greater lipid accumulation in the liver, and show deterioration of hepatic insulin signaling. The worsening of metabolic dysfunction is also influenced, in part, by the length of the diet exposure, as these metabolic parameters worsened between ten and twenty weeks on the diets. In addition to the baseline diet and length of dietary exposure, the response to fructose intake is driven by sex differences. While fructose-fed male mice develop metabolic dysfunction, female mice fed fructose for twenty weeks do not gain weight and remain insulin-sensitive despite accumulating a large amount of lipids in the liver. Together, these findings suggest that fructose-induced metabolic dysfunction is not a universal outcome but is highly dependent on the baseline diet, the length of the dietary exposure, and sex.

Our findings indicate that the baseline diet strongly modulates the metabolic response to fructose. In terms of weight gain, fructose supplementation resulted in significant weight gain on BCD; there was no weight gain on LXD, and mice tended to weigh less when given fructose on LFD. Moreover, the same mice that did not gain weight when fed fructose on LFD gained weight

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

when the baseline diet was switched to BCD. The response to fructose supplementation could be, in part, explained by the carbohydrate load of the baseline diet. When mice consume fructose together with a carbohydrate-rich LFD, their solid food intake decreases the most. Solid food intake is also lower with fructose consumption on the LXD, which contains fewer carbohydrates than the LFD, but the reduction is not as pronounced as on the LFD. Interestingly, there was no reduction in solid food intake when fructose was consumed on the BCD, which is relatively high in fat. Therefore, the carbohydrate load in the solid diet is critical in determining how liquid fructose affects the intake of the solid diet. Indeed, Solon-Biet et al. found that sweet preference for sugar water depends on carbohydrate balance, whereas mice on a high carbohydrate diet drink less sucrose and mice on a low carbohydrate diet drink more sucrose [35]. The studies of nutritional geometry also indicate that the protein content of the diet drives energy intake from a solid diet [36]. All three of our diets, BCD, LXD, and LFD, contain similar amounts of protein (23, 24, and 20% protein), so it's unlikely that the protein content affected fructose-adjusted solid food intake. If anything, the mice prefer the average protein intake to be 23% [36], so it would be expected that they would eat more, not less, of LFD to keep the optimal protein intake when supplemented with fructose. In summary, drinking fructose on a low-fat, high-carbohydrate diet triggers a compensatory decrease in solid food intake and prevents the development of metabolic complications. This is akin to the protective effects of fructose when consumed from fruits and vegetables as a part of a well-balanced low-fat diet, while consumption of sugar-sweetened beverages on a standard American diet is harmful to metabolic health [38].

Fructose metabolism via KHK-C strongly induces lipogenesis [39, 40]. Our results demonstrate that the baseline diet also influences the propensity of fructose to induce hepatic steatosis. Indeed, fructose induces hepatic triglyceride accumulation when consumed on a BCD, but not on a LXD or LFD. These differences can be explained by the propensity of the

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

baseline diet to suppress KHK-C and endogenous DNL pathway. Indeed, diets high in fat lower endogenous DNL [41]. The BCD has the highest amount of fat and, therefore, the lowest percentage of carbohydrates. Thus, the baseline levels of KHK-C and DNL proteins were the lowest on BCD, and they experienced the greatest fold-induction with fructose supplementation. On the other hand, the baseline protein levels of KHK-C and DNL enzymes were the highest on LFD since this diet has the lowest fat content. Upon addition of fructose to LFD, there is only a marginal increase in KHK-C and DNL enzymes. Thus, fructose increases KHK-C and endogenous DNL pathway in all diets, but the diets that have high-fat content and, therefore, low baseline lipogenesis exhibit the greatest fold-increase in lipogenesis with fructose supplementation. On the other hand, fructose supplementation also reduces the protein levels of CPT1A, the rate-limiting enzyme of FAO, in all three diets. Since the BCD contains the highest percent of calories from fat, it is understandable that fructose intake would have the largest effect on this diet. These findings support the well-established role of fructose in reducing FAO [42]. Another interesting aspect of our research documents that fructose intake had a larger effect on the protein levels of DNL and FAO enzymes rather than on their mRNA expression. This observation is in line with our prior studies documenting that fructose via KHK-C affects the post-translational protein modifications and, thus, protein stability [30, 31]. Taken together, our results show that consumption of fructose along with a diet that contains the largest amount of fat supports the development of hepatic steatosis. While fructose induces KHK-C and hepatic steatosis in all conditions tested, interestingly, steatosis does not always correlate with weight gain or hepatic insulin resistance. This makes targeting KHK-C for the management of Metabolic dysfunction-associated steatotic liver disease (MASLD) attractive for patients with or without obesity and in both sexes.

Sex is another biological variable that profoundly affects the propensity of fructose to induce metabolic complications. Male mice supplemented with fructose gain weight, develop

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

impaired glucose tolerance, and hepatic steatosis. Female mice, on the other hand, do not gain additional weight and remain glucose-tolerant when exposed to fructose. This is in agreement with a generally lower risk of insulin resistance [43] and fatty liver disease [44, 45] in women compared to men. Even though female mice do not gain additional weight when exposed to fructose, they have a larger induction in KHK-C and develop profound hepatic steatosis. This indicates that female mice are better able to compensate for the detrimental effects of fructose rather than that they are protected from developing fructose-induced complications. Indeed, the propensity of female mice to gain weight increases following ovariectomy [46] or housing at thermoneutral environment [47]. Thus, it remains to be determined if fructose intake in female mice could lead to full-blown metabolic complications when combined with other metabolic risk factors.

The limitations of this study include utilizing pure fructose-sweetened drinks when the majority of SSBs are sweetened with sucrose or high-fructose corn syrup, which is a mixture of fructose and glucose. Indeed, Wali et al., reported that the worst metabolic outcomes occur when drinks contain a 50:50 fructose to glucose mixture [48, 49]. Moreover, FGF21 is robustly increased with sugar intake, while exogenous FGF21 administration reduces sugar consumption [50, 51]. We did not investigate FGF21 biology in our study. Lastly, the microbiome metabolizes excess fructose to acetate, which circulates to the liver to support DNL [37]. The limitation of our study is that we did not investigate the microbiome effects on the propensity of fructose to induce metabolic dysfunction on different diets.

5. Conclusions

This study demonstrates that the metabolic effects of fructose are highly dependent on baseline diet, length of dietary exposure, and sex as a biological variable. Under all conditions tested, fructose intake increased KHK-C and activated the endogenous DNL pathway. However, weight gain and hepatic insulin resistance did not universally follow high fructose intake. These

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

findings highlight the complexity of fructose metabolism on metabolic health, emphasizing the need to consider both the underlying diet and patient-specific risk factors.

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

Authorship Contributions Statement

Conceptualization, T.F. and S.S.; investigation, T.F., S.P., H.A., J.A.W. and S.S; data curation, T.F. and S.S; formal analysis, T.F. and S.S.; methodology, T.F., S.P., H.A., J.A.W. and S.S; writing – original draft, T.F. and S.S.; writing – review & editing, T.F., S.P., H.A., Y.A., J.A.W., S.J.S. and S.S; funding acquisition, S.S.; project administration, S.S.; resources, S.S.; software, T.F. and S.S.; supervision, S.S.; validation, S.S.; visualization, S.S. All authors have read and agreed to the published version of the manuscript.

Funding

This work was supported in part by the NASPGHAN Foundation Young Investigator Award, the Pediatric Scientist Development Program Award (HD000850), and the COCVD Pilot and Feasibility Grant (GM127211) awarded to SS. This research was supported by the University of Kentucky Office of the Vice President for Research through the Diabetes and Obesity Research Priority Area.

Institutional Review Board Statement

In accordance with NIH guidelines, all animal protocols were approved by the IACUC of the University of Kentucky, and Joslin Diabetes Center, where different experiments were performed.

Conflicts of Interest

The authors declare no conflict of interest. J.A.W. is now employed by Nestlé Research. Nestlé neither funded this work nor provided any input into this manuscript.

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

FIGURE LEGENDS

Figure 1. The baseline diet modifies the impact of fructose on weight gain and insulin resistance. A) Weight gain of male mice on Boston Chow Diet (9F5020) supplemented with either regular water (BCD+W) or 30% fructose-sweetened water (BCD+F), for 10 weeks, (n=8 mice per group). **B)** Weight gain of male mice on Lexington Chow Diet (T2018) supplemented with either regular water (LXD+W) or 30% fructose-sweetened water (LXD+F) for 10 weeks, n=8 mice per group. **C)** Average total caloric intake in mice on BCD, **D)** and LXD. **E)** Fasting insulin levels, **F)** blood glucose concentrations, and **G)** Homeostatic Model Assessment for Insulin Resistance (HOMA-IR) for BCD-fed mice at 8 weeks on diet. **H)** Fasting insulin levels, **I)** blood glucose concentrations, and **J)** HOMA-IR for LXD-fed mice at 8 weeks on the diet. **K)** Glucose tolerance test (GTT) for BCD-fed mice performed after 8 weeks and **L)** Area under the curve (AUC) analysis for GTT results in BCD-fed mice. **M)** Insulin tolerance test (ITT) in BCD-fed mice at 8 weeks and **N)** Area above the curve (AAC) for ITT in BCD-fed mice. **O)** GTT results for LXD-fed mice at 8 weeks and **P)** AUC analysis for GTT. **Q)** ITT in LXD-fed mice at 8 weeks on the diet and **R)** AAC for ITT in LXD-fed mice. **S)** Western blot analysis of phosphorylated and total AKT and ERK proteins in BCD and LXD-fed mice (n=4 mice/group), and **T)** Image J quantification of Western blot results for phosphorylated and total AKT and ERK proteins. Statistical comparisons were conducted using Student's t-test for pairwise comparisons between the control and fructose groups. For analyses involving multiple groups, i.e., diet and fructose, two-way ANOVA followed by Tukey's post hoc test was performed. Statistical significance is indicated by asterisks (*) when comparing fructose-supplemented groups to their respective controls: *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$. Data are presented as mean ± SEM.

Figure 2. Fructose supplementation on BCD, but not on LXD, supports hepatic steatosis. A) Liver weights of BCD-fed mice at sacrifice after 10 weeks on the diet. **B)** Liver weights of

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

LXD-fed mice at sacrifice after 10 weeks on the diet. **C)** Oil Red O staining of liver sections from BCD and LXD-fed mice, showing hepatic lipid accumulation. **D)** Hepatic triglyceride (TG) content in BCD-fed and **E)** LXD-fed mice given fructose or regular water. **F)** mRNA expression of KHK isoforms in the liver of BCD-fed, **G)** and LXD-fed mice. **H)** Protein levels of KHK isoforms in the liver of BCD and LXD-fed mice, respectively. **I)** Hepatic mRNA expression of enzymes involved in fatty acid synthesis in BCD-fed, **J)** and LXD-fed mice. n=7-8 mice per group. **K)** Protein levels of enzymes involved in hepatic fatty acid synthesis in the livers of BCD and LXD-fed mice, respectively. **L)** Protein levels of enzymes involved in hepatic fatty acid oxidation in BCD and LXD-fed mice. (n=4 mice/group). Statistical comparisons were conducted using Student's t-test for pairwise comparisons between the control and fructose groups. Statistical significance is indicated by asterisks (*) when comparing fructose-supplemented groups to their respective controls: *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$. Data are presented as mean ± SEM.

Figure 3. Switching from a low-fat diet to a Boston chow diet restores weight gain and the development of insulin resistance. A) Weight gain of male mice on Low Fat Diet (D12450K) provided with either regular water (LFD+W) or 30% fructose-sweetened water (LFD+F) for 10 weeks, n=8 mice per group. **B)** Weight gain of male mice switched to the Boston Chow Diet, supplemented with either regular water (BCD+W) or 30% fructose-sweetened water (BCD+F) for an additional 10 weeks, n=8 mice per group. **C)** Total caloric intake in LFD-fed, **D)** and BCD-fed mice. **E)** Fasting insulin levels, **F)** blood glucose concentrations, and **G)** HOMA-IR in LFD-fed mice measured after 10 weeks on the diet. **H)** Fasting insulin levels, **I)** blood glucose concentrations, and **J)** HOMA-IR for BCD-fed mice at 20 weeks on the diet. **K)** A glucose tolerance test (GTT) was performed in LFD-fed mice after 8 weeks on the diet and **L)** AUC for GTT results in LFD-fed mice. **M)** Insulin tolerance test (ITT) performed in LFD-fed mice at 8 weeks on the diet and **N)** Area above the curve (AAC) for ITT in LFD-fed mice. **O)** GTT results

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

for BCD-fed mice at 18 weeks on the diet and **P)** AUC analysis for GTT results in BCD-fed mice. **Q)** ITT results for BCD-fed mice at 18 weeks on the diet and **R)** AAC analysis for ITT in BCD-fed mice. **S)** Western blot (WB) analysis of phosphorylated and total AKT and ERK proteins in LFD- and BCD-fed mice, respectively, and **T)** Image J quantification showing protein levels of phosphorylated and total AKT and ERK (n=4 mice/group). Statistical comparisons were conducted using Student's t-test for pairwise comparisons between the control and fructose groups. For analyses involving multiple groups, i.e., diet and fructose, two-way ANOVA followed by Tukey's post hoc test was performed. Statistical significance is indicated by asterisks (*) when comparing fructose-supplemented groups to their respective controls: *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$. Data are presented as mean ± SEM.

Figure 4. Fructose enhances hepatic steatosis more strongly on BCD than on LFD. A) Liver weights of LFD-fed mice at sacrifice after 10 weeks on the diet. **B)** Liver weights of BCD-fed mice at sacrifice after 20 weeks on the diet. **C)** Oil Red O staining of liver sections from LFD- and BCD-fed mice, highlighting hepatic lipid accumulation. **D)** Hepatic triglyceride (TG) content in LFD-fed, **E)** and BCD-fed mice. **F)** mRNA expression of KHK isoforms in the liver of LFD-fed, **G)** and BCD-fed mice. **H)** Protein levels of KHK isoforms in the liver of LFD- and BCD-fed mice. **I)** mRNA expression of enzymes involved in hepatic fatty acid synthesis in LFD-fed, **J)** and BCD-fed mice. n=6 mice per group. **K)** Protein levels of enzymes involved in hepatic fatty acid synthesis in the liver of LFD and BCD-fed mice. **L)** Protein levels of enzymes involved in hepatic fatty acid oxidation in LFD and BCD-fed mice (n=4 mice/group). Statistical comparisons were conducted using Student's t-test for pairwise comparisons between the control and fructose groups. Statistical significance is indicated by asterisks (*) when comparing fructose-supplemented groups to their respective controls: *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$. Data are presented as mean ± SEM.

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

Figure 5. Fructose supplementation results in weight gain and insulin resistance in male but not female mice. A) Weight gain of male and female mice on LFD for 10 weeks then switched over to BCD for another 10 weeks and supplemented with either regular water or 30% fructose-sweetened water (n=16 mice/group for first 10 weeks and n=8 mice/group for the next 10 weeks). **B)** Water consumption, **C)** food consumption, and **D)** total calorie intake of male and female mice from weeks 11 to 20 on the diets. **E)** Fasting insulin levels, **F)** blood glucose concentrations, and **G)** HOMA-IR in male and female mice after 20 weeks on the diet. **H)** A glucose tolerance test (GTT) results, **I)** and area under the curve (AUC) analysis for male and female mice, performed after 18 weeks on the diet. **J)** An insulin tolerance test (ITT), and **K)** area above the curve (AAC) analysis for male and female mice, performed after 18 weeks on the diet. **L)** Western blot analysis (WB) of phosphorylated and total AKT and ERK proteins in liver tissues of male and female mice, respectively, and **M)** Image J quantification showing protein levels of phosphorylated and total AKT and ERK (n=4 mice/group). Statistical analysis was performed using two-way ANOVA followed by Tukey's post hoc test for comparisons across multiple groups. Statistical significance, denoted by asterisks (*), represents comparisons of fructose-supplemented groups to their respective controls, as follows: *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$. All data are expressed as mean ± SEM.

Figure 6. Fructose supplementation in female mice more strongly upregulates KHK-C and DNL enzymes than in males. A) Liver weights of male and female mice at the time of sacrifice after 20 weeks on the diet. **B)** Liver-to-body weight ratio in male and female mice. **C)** Oil Red O staining of liver sections from male and female mice, showing lipid accumulation. **D)** Hepatic triglyceride (TG) content, **E)** and liver cholesterol content in male and female mice after 20 weeks on the diet. **F)** Protein levels of KHK isoforms in the liver of male and female mice. **G)** Protein levels of enzymes involved in the hepatic fatty acid synthesis, **H)** and ImageJ quantification of these proteins in the livers of male and female mice. **I)** Protein levels of

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

enzymes involved in hepatic fatty acid oxidation, **J)** and ImageJ quantification of these proteins in male and female mice (n=4 mice/group). Statistical analysis was performed using two-way ANOVA followed by Tukey's post hoc test for comparisons across multiple groups. Statistical significance, denoted by asterisks (*), represents comparisons of fructose-supplemented groups to their respective controls, as follows: *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$. All data are expressed as mean ± SEM.

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

References

1. Ford, E.S., W.H. Giles, and W.H. Dietz, *Prevalence of the metabolic syndrome among US adults: findings from the third National Health and Nutrition Examination Survey.* JAMA, 2002. **287**(3): p. 356-9.
2. Magkos, F., et al., *On the pathogenesis of obesity: causal models and missing pieces of the puzzle.* Nat Metab, 2024. **6**(10): p. 1856-1865.
3. Gillespie, K.M., et al., *The Impact of Free Sugar on Human Health-A Narrative Review.* Nutrients, 2023. **15**(4).
4. Bantle, J.P., *Dietary fructose and metabolic syndrome and diabetes.* J Nutr, 2009. **139**(6): p. 1263S-1268S.
5. Stanhope, K.L., *Role of fructose-containing sugars in the epidemics of obesity and metabolic syndrome.* Annu Rev Med, 2012. **63**: p. 329-43.
6. Warshaw, H. and S.V. Edelman, *Practical Strategies to Help Reduce Added Sugars Consumption to Support Glycemic and Weight Management Goals.* Clin Diabetes, 2021. **39**(1): p. 45-56.
7. Cohen, C.C., et al., *Dietary sugar restriction reduces hepatic de novo lipogenesis in adolescent boys with fatty liver disease.* J Clin Invest, 2021. **131**(24).
8. Radulescu, A., et al., *Dietary Counseling Aimed at Reducing Sugar Intake Yields the Greatest Improvement in Management of Weight and Metabolic Dysfunction in Children with Obesity.* Nutrients, 2022. **14**(7).
9. Lustig, R.H., et al., *Isocaloric fructose restriction and metabolic improvement in children with obesity and metabolic syndrome.* Obesity (Silver Spring), 2016. **24**(2): p. 453-60.
10. Tappy, L. and K.A. Le, *Metabolic effects of fructose and the worldwide increase in obesity.* Physiol Rev, 2010. **90**(1): p. 23-46.
11. Hartley, L., et al., *Increased consumption of fruit and vegetables for the primary prevention of cardiovascular diseases.* Cochrane Database Syst Rev, 2012. **2012**(5).
12. Malik, V.S., et al., *Sugar-sweetened beverages and risk of metabolic syndrome and type 2 diabetes: a meta-analysis.* Diabetes Care, 2010. **33**(11): p. 2477-83.
13. Flood-Obbagy, J.E. and B.J. Rolls, *The effect of fruit in different forms on energy intake and satiety at a meal.* Appetite, 2009. **52**(2): p. 416-22.
14. Merino, B., et al., *Intestinal Fructose and Glucose Metabolism in Health and Disease.* Nutrients, 2019. **12**(1).
15. Jang, C., et al., *The Small Intestine Converts Dietary Fructose into Glucose and Organic Acids.* Cell Metab, 2018. **27**(2): p. 351-361 e3.
16. Jang, C., et al., *The small intestine shields the liver from fructose-induced steatosis.* Nat Metab, 2020. **2**(7): p. 586-593.
17. Holt, E.M., et al., *Fruit and vegetable consumption and its relation to markers of inflammation and oxidative stress in adolescents.* J Am Diet Assoc, 2009. **109**(3): p. 414-21.
18. Stanhope, K.L. and P.J. Havel, *Endocrine and metabolic effects of consuming beverages sweetened with fructose, glucose, sucrose, or high-fructose corn syrup.* Am J Clin Nutr, 2008. **88**(6): p. 1733S-1737S.
19. Elliott, S.S., et al., *Fructose, weight gain, and the insulin resistance syndrome.* Am J Clin Nutr, 2002. **76**(5): p. 911-22.
20. Tordoff, M.G. and A.M. Alleva, *Effect of drinking soda sweetened with aspartame or high-fructose corn syrup on food intake and body weight.* Am J Clin Nutr, 1990. **51**(6): p. 963-9.
21. Malik, V.S., M.B. Schulze, and F.B. Hu, *Intake of sugar-sweetened beverages and weight gain: a systematic review.* Am J Clin Nutr, 2006. **84**(2): p. 274-88.

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

22. van Buul, V.J., L. Tappy, and F.J. Brouns, *Misconceptions about fructose-containing sugars and their role in the obesity epidemic.* Nutr Res Rev, 2014. **27**(1): p. 119-30.
23. Lowndes, J., et al., *The effect of normally consumed amounts of sucrose or high fructose corn syrup on lipid profiles, body composition and related parameters in overweight/obese subjects.* Nutrients, 2014. **6**(3): p. 1128-44.
24. Hallfrisch, J., et al., *Effects of dietary fructose on plasma glucose and hormone responses in normal and hyperinsulinemic men.* J Nutr, 1983. **113**(9): p. 1819-26.
25. Matikainen, N., et al., *Fructose intervention for 12 weeks does not impair glycemic control or incretin hormone responses during oral glucose or mixed meal tests in obese men.* Nutr Metab Cardiovasc Dis, 2017. **27**(6): p. 534-542.
26. Lowndes, J., S.S. Sinnett, and J.M. Rippe, *No Effect of Added Sugar Consumed at Median American Intake Level on Glucose Tolerance or Insulin Resistance.* Nutrients, 2015. **7**(10): p. 8830-45.
27. Crapo, P.A., O.G. Kolterman, and J.M. Olefsky, *Effects of oral fructose in normal, diabetic, and impaired glucose tolerance subjects.* Diabetes Care, 1980. **3**(5): p. 575-82.
28. Johnson, R.K., et al., *Dietary sugars intake and cardiovascular health: a scientific statement from the American Heart Association.* Circulation, 2009. **120**(11): p. 1011-20.
29. Canadian Diabetes Association Clinical Practice Guidelines Expert, C., et al., *Nutrition therapy.* Can J Diabetes, 2013. **37 Suppl 1**: p. S45-55.
30. Softic, S., et al., *Dietary Sugars Alter Hepatic Fatty Acid Oxidation via Transcriptional and Post-translational Modifications of Mitochondrial Proteins.* Cell Metab, 2019. **30**(4): p. 735-753 e4.
31. Helsley, R.N., et al., *Ketohexokinase-C regulates global protein acetylation to decrease carnitine palmitoyltransferase 1a-mediated fatty acid oxidation.* J Hepatol, 2023. **79**(1): p. 25-42.
32. Park, S.H., et al., *Fructose induced KHK-C can increase ER stress independent of its effect on lipogenesis to drive liver disease in diet-induced and genetic models of NAFLD.* Metabolism, 2023. **145**: p. 155591.
33. Diggle, C.P., et al., *Ketohexokinase: expression and localization of the principal fructose-metabolizing enzyme.* J Histochem Cytochem, 2009. **57**(8): p. 763-74.
34. Softic, S., D.E. Cohen, and C.R. Kahn, *Role of Dietary Fructose and Hepatic De Novo Lipogenesis in Fatty Liver Disease.* Dig Dis Sci, 2016. **61**(5): p. 1282-93.
35. Solon-Biet, S.M., et al., *Toward reconciling the roles of FGF21 in protein appetite, sweet preference, and energy expenditure.* Cell Rep, 2023. **42**(12): p. 113536.
36. Sorensen, A., et al., *Protein-leverage in mice: the geometry of macronutrient balancing and consequences for fat deposition.* Obesity (Silver Spring), 2008. **16**(3): p. 566-71.
37. Zhao, S., et al., *Dietary fructose feeds hepatic lipogenesis via microbiota-derived acetate.* Nature, 2020. **579**(7800): p. 586-591.
38. Li, X., et al., *Fructose consumption from different food sources and cardiometabolic biomarkers: cross-sectional associations in US men and women.* Am J Clin Nutr, 2023. **117**(3): p. 490-498.
39. Ishimoto, T., et al., *High-fat and high-sucrose (western) diet induces steatohepatitis that is dependent on fructokinase.* Hepatology, 2013. **58**(5): p. 1632-43.
40. Softic, S., et al., *Divergent effects of glucose and fructose on hepatic lipogenesis and insulin signaling.* J Clin Invest, 2017. **127**(11): p. 4059-4074.
41. Duarte, J.A., et al., *A high-fat diet suppresses de novo lipogenesis and desaturation but not elongation and triglyceride synthesis in mice.* J Lipid Res, 2014. **55**(12): p. 2541-53.
42. Inci, M.K., et al., *Fructose Impairs Fat Oxidation: Implications for the Mechanism of Western diet-induced NAFLD.* J Nutr Biochem, 2022: p. 109224.
43. Geer, E.B. and W. Shen, *Gender differences in insulin resistance, body composition, and energy balance.* Gend Med, 2009. **6 Suppl 1**(Suppl 1): p. 60-75.

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

44. Lazo, M., et al., *Prevalence of nonalcoholic fatty liver disease in the United States: the Third National Health and Nutrition Examination Survey, 1988-1994.* Am J Epidemiol, 2013. **178**(1): p. 38-45.
45. Radulescu, A., et al., *Stratification by obesity class, rather than age, can identify a higher percent of children at risk for non-alcoholic fatty liver disease and metabolic dysfunction.* Pediatr Obes, 2021: p. e12862.
46. Rogers, N.H., et al., *Reduced energy expenditure and increased inflammation are early events in the development of ovariectomy-induced obesity.* Endocrinology, 2009. **150**(5): p. 2161-8.
47. Giles, D.A., et al., *Thermoneutral housing exacerbates nonalcoholic fatty liver disease in mice and allows for sex-independent disease modeling.* Nat Med, 2017. **23**(7): p. 829-838.
48. Wali, J.A., et al., *Impact of dietary carbohydrate type and protein-carbohydrate interaction on metabolic health.* Nat Metab, 2021. **3**(6): p. 810-828.
49. Wali, J.A., et al., *Determining the metabolic effects of dietary fat, sugars and fat-sugar interaction using nutritional geometry in a dietary challenge study with male mice.* Nat Commun, 2023. **14**(1): p. 4409.
50. Solon-Biet, S.M., et al., *Defining the Nutritional and Metabolic Context of FGF21 Using the Geometric Framework.* Cell Metab, 2016. **24**(4): p. 555-565.
51. Hill, C.M., et al., *FGF21 and the Physiological Regulation of Macronutrient Preference.* Endocrinology, 2020. **161**(3).

Disclaimer/Publisher's Note: The statements, opinions, and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions, or products referred to in the content.

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

Figure 1

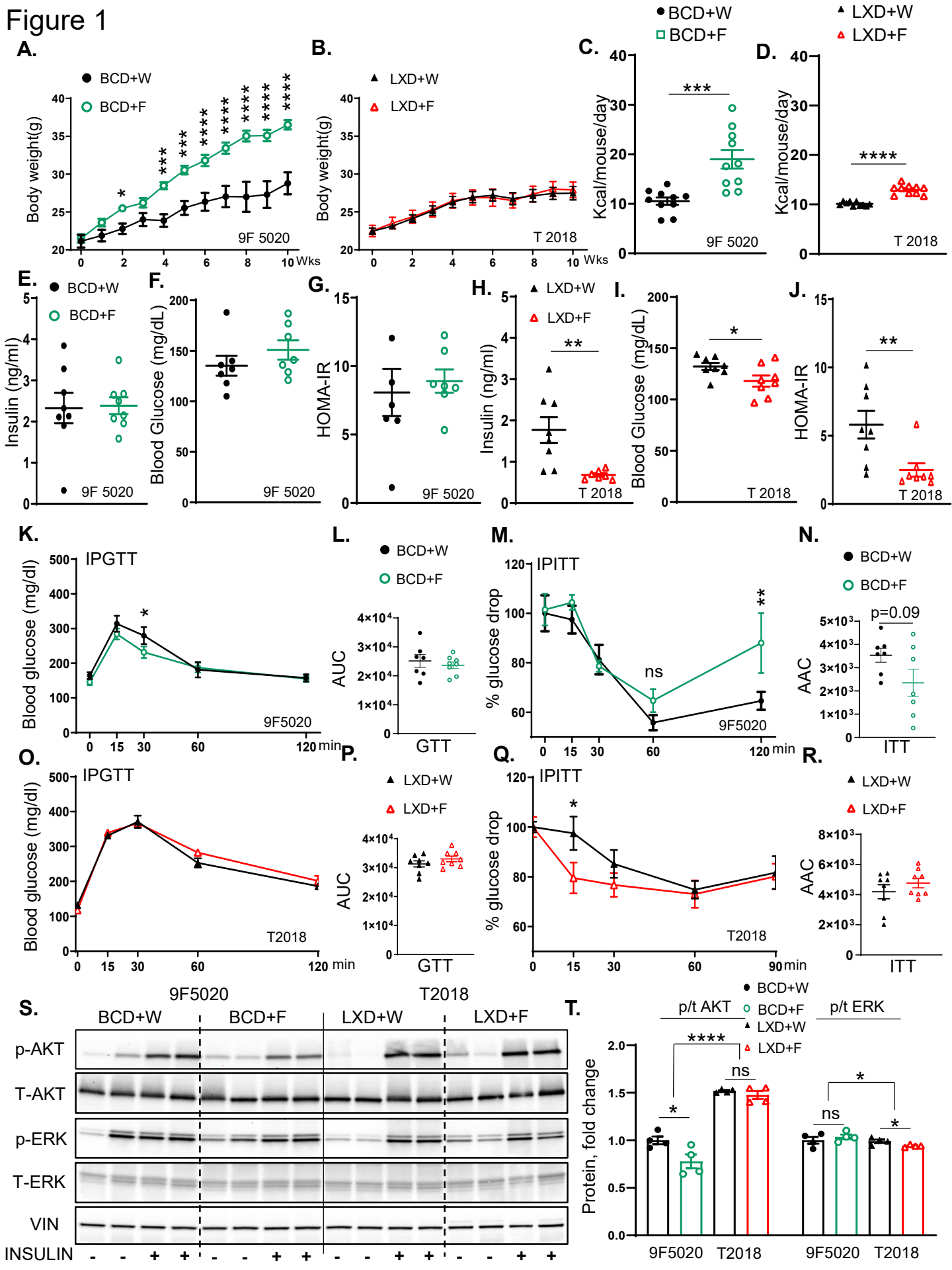

A.
BCD+W
BCD+F
Body weight(g)
9F 5020
Wks
B.
LXD+W
LXD+F
Body weight(g)
T 2018
Wks
C.
BCD+W
BCD+F
Kcal/mouse/day
9F 5020
D.
LXD+W
LXD+F
Kcal/mouse/day
T 2018
E.
BCD+W
BCD+F
Insulin (ng/ml)
9F 5020
F.
Blood Glucose (mg/dL)
9F 5020
G.
HOMA-IR
9F 5020
H.
LXD+W
LXD+F
Insulin (ng/ml)
T 2018
I.
Blood Glucose (mg/dL)
T 2018
J.
HOMA-IR
T 2018
K.
IPGTT
Blood glucose (mg/dl)
9F5020
min
L.
BCD+W
BCD+F
AUC
GTT
M.
IPITT
% glucose drop
ns
9F5020
min
N.
BCD+W
BCD+F
p=0.09
AAC
ITT
O.
IPGTT
Blood glucose (mg/dl)
T2018
min
P.
LXD+W
LXD+F
AUC
GTT
Q.
IPITT
% glucose drop
T2018
min
R.
LXD+W
LXD+F
AAC
ITT
S.
9F5020
T2018
BCD+W
BCD+F
LXD+W
LXD+F
p-AKT
T-AKT
p-ERK
T-ERK
VIN
INSULIN
T.
BCD+W
BCD+F
LXD+W
LXD+F
p/t AKT
p/t ERK
Protein, fold change
ns
9F5020
T2018
9F5020
T2018

Figure 2

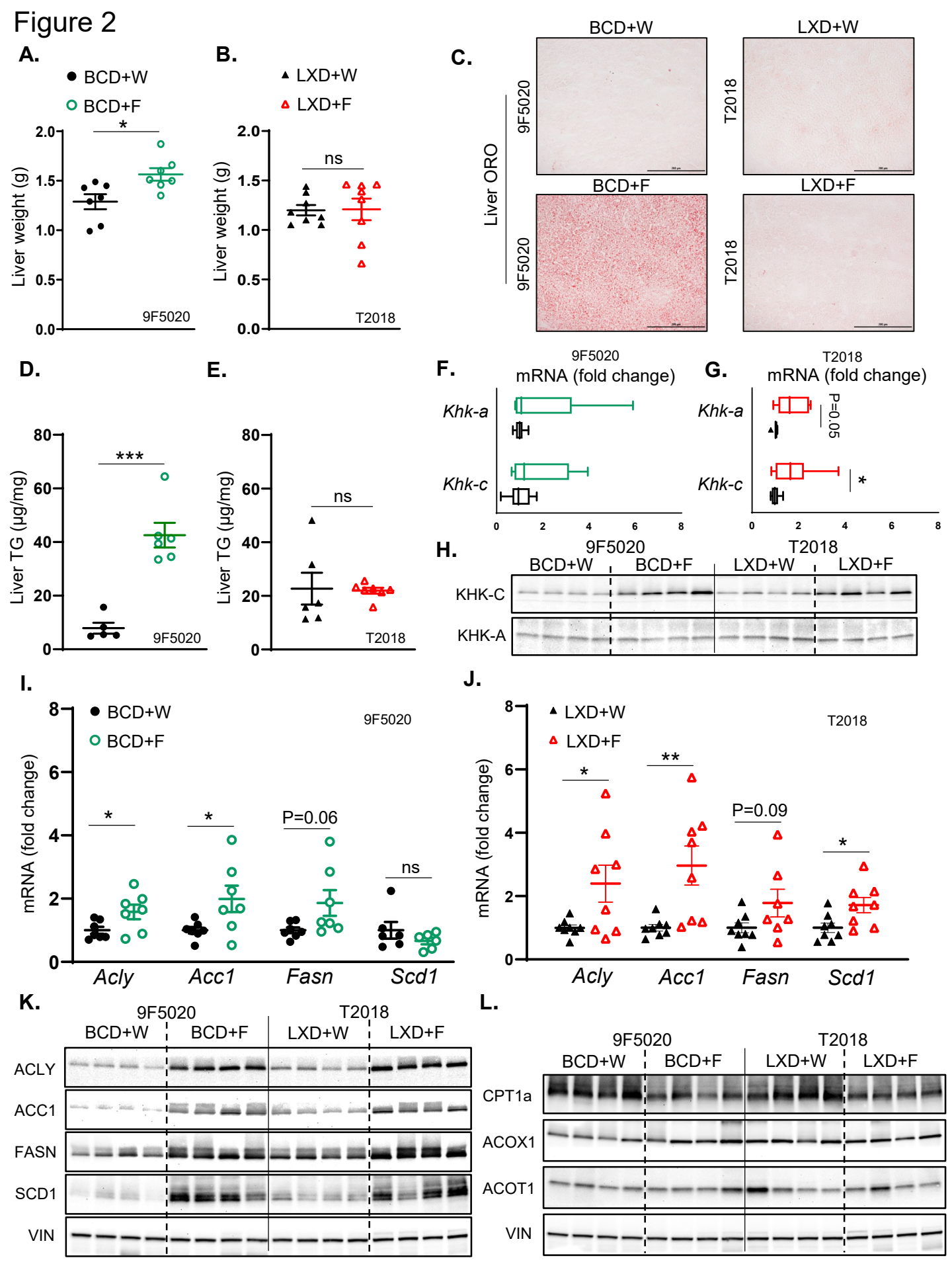

A.
BCD+W
BCD+F
*
Liver weight (g)
2.0
1.5
1.0
0.5
0.0
9F5020
B.
LXD+W
LXD+F
ns
Liver weight (g)
T2018
C.
BCD+W
LXD+W
BCD+F
LXD+F
Liver ORO
9F5020
T2018
D.
***
Liver TG (µg/mg)
80
60
40
20
0
9F5020
E.
ns
Liver TG (µg/mg)
T2018
F.
9F5020
mRNA (fold change)
Khk-a
Khk-c
0
2
4
6
8
G.
T2018
mRNA (fold change)
Khk-a
Khk-c
P=0.05
*
H.
9F5020
T2018
BCD+W
BCD+F
LXD+W
LXD+F
KHK-C
KHK-A
I.
BCD+W
BCD+F
9F5020
mRNA (fold change)
*
*
P=0.06
ns
Acly
Acc1
Fasn
Scd1
J.
LXD+W
LXD+F
T2018
mRNA (fold change)
*
**
P=0.09
*
Acly
Acc1
Fasn
Scd1
K.
9F5020
T2018
BCD+W
BCD+F
LXD+W
LXD+F
ACLY
ACC1
FASN
SCD1
VIN
L.
9F5020
T2018
BCD+W
BCD+F
LXD+W
LXD+F
CPT1a
ACOX1
ACOT1
VIN

Figure 3

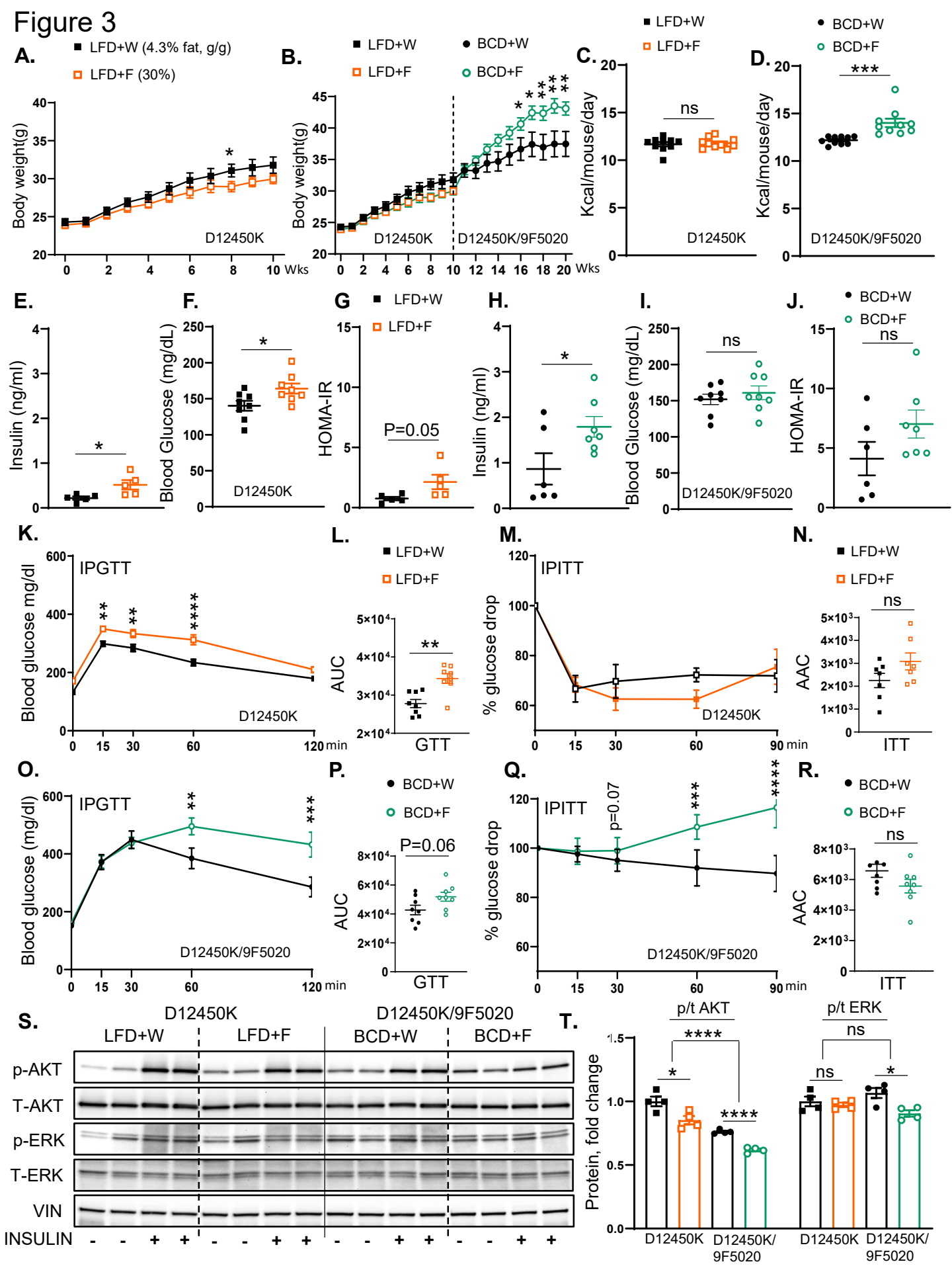

A.
LFD+W (4.3% fat, g/g)
LFD+F (30%)
Body weight(g)
D12450K
Wks
B.
LFD+W
LFD+F
BCD+W
BCD+F
Body weight(g)
D12450K
D12450K/9F5020
Wks
C.
LFD+W
LFD+F
Kcal/mouse/day
ns
D12450K
D.
BCD+W
BCD+F
Kcal/mouse/day
***
D12450K/9F5020
E.
Insulin (ng/ml)
F.
Blood Glucose (mg/dL)
D12450K
G
LFD+W
LFD+F
HOMA-IR
P=0.05
H.
Insulin (ng/ml)
I.
Blood Glucose (mg/dL)
ns
D12450K/9F5020
J.
BCD+W
BCD+F
HOMA-IR
ns
K.
IPGTT
Blood glucose mg/dl
D12450K
min
L.
LFD+W
LFD+F
AUC
GTT
M.
IPITT
% glucose drop
D12450K
min
N.
LFD+W
LFD+F
ns
AAC
ITT
O.
IPGTT
Blood glucose (mg/dl)
D12450K/9F5020
min
P.
BCD+W
BCD+F
P=0.06
AUC
GTT
Q.
IPITT
p=0.07
% glucose drop
D12450K/9F5020
min
R.
BCD+W
BCD+F
ns
AAC
ITT
S.
D12450K
D12450K/9F5020
LFD+W
LFD+F
BCD+W
BCD+F
p-AKT
T-AKT
p-ERK
T-ERK
VIN
INSULIN
T.
p/t AKT
p/t ERK
ns
Protein, fold change
D12450K
D12450K/9F5020
D12450K
D12450K/9F5020

Figure 4

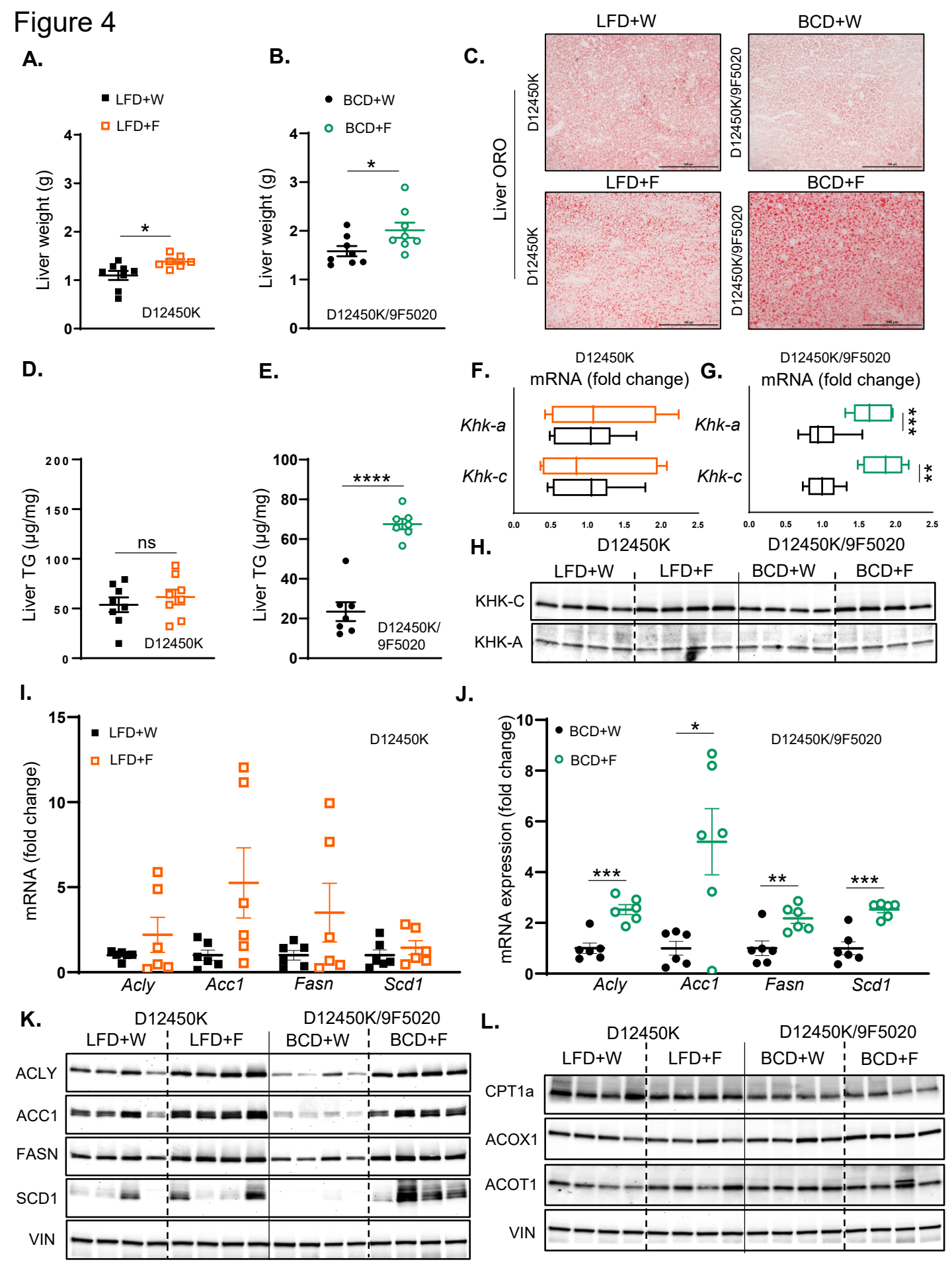

A.
LFD+W
LFD+F
Liver weight (g)
*
D12450K
B.
BCD+W
BCD+F
Liver weight (g)
*
D12450K/9F5020
C.
LFD+W
BCD+W
LFD+F
BCD+F
Liver ORO
D12450K
D12450K/9F5020
D12450K
D12450K/9F5020
D.
Liver TG (µg/mg)
ns
D12450K
E.
Liver TG (µg/mg)
****
D12450K/
9F5020
F.
D12450K
mRNA (fold change)
Khk-a
Khk-c
G.
D12450K/9F5020
mRNA (fold change)
Khk-a
Khk-c
***
**
H.
D12450K
D12450K/9F5020
LFD+W
LFD+F
BCD+W
BCD+F
KHK-C
KHK-A
I.
LFD+W
LFD+F
D12450K
mRNA (fold change)
Acly
Acc1
Fasn
Scd1
J.
BCD+W
BCD+F
D12450K/9F5020
mRNA expression (fold change)
*
***
**
***
Acly
Acc1
Fasn
Scd1
K.
D12450K
D12450K/9F5020
LFD+W
LFD+F
BCD+W
BCD+F
ACLY
ACC1
FASN
SCD1
VIN
L.
D12450K
D12450K/9F5020
LFD+W
LFD+F
BCD+W
BCD+F
CPT1a
ACOX1
ACOT1
VIN

Figure 5

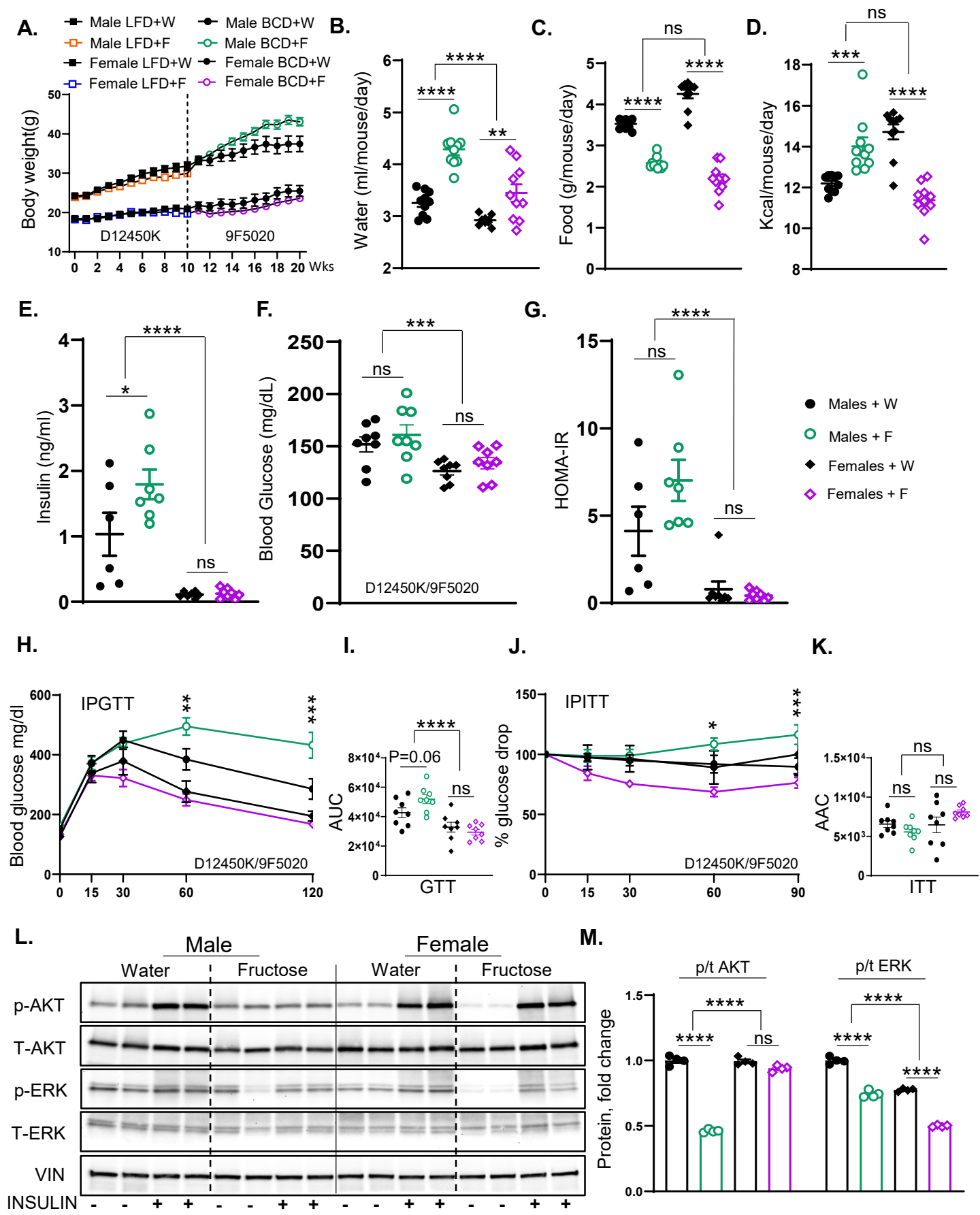
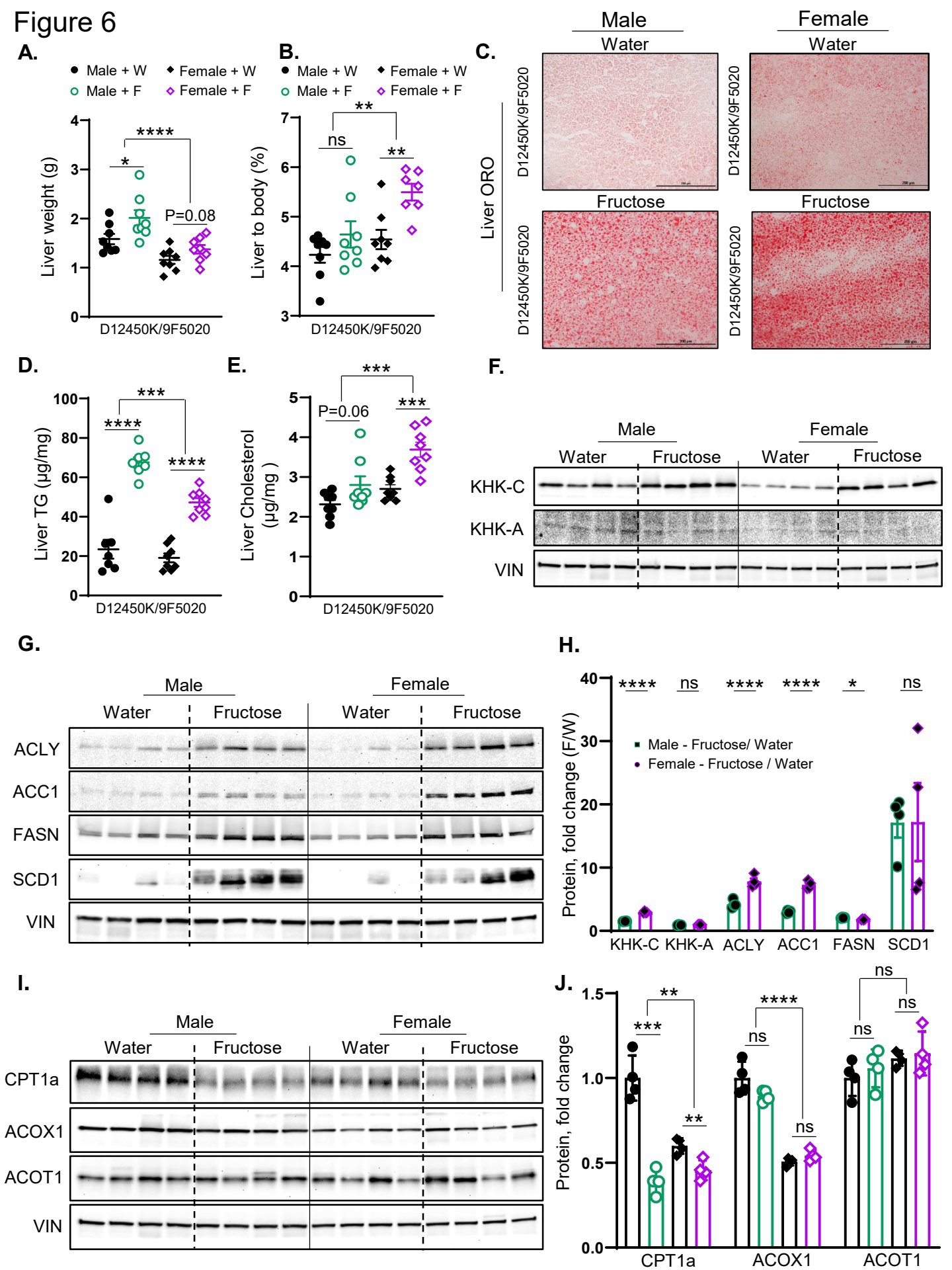

A.
Male LFD+W
Male LFD+F
Female LFD+W
Female LFD+F
Male BCD+W
Male BCD+F
Female BCD+W
Female BCD+F
Body weight(g)
D12450K
9F5020
Wks
B.
Water (ml/mouse/day)
C.
Food (g/mouse/day)
D.
Kcal/mouse/day
E.
Insulin (ng/ml)
F.
Blood Glucose (mg/dL)
D12450K/9F5020
G.
HOMA-IR
Males + W
Males + F
Females + W
Females + F
H.
IPGTT
Blood glucose mg/dl
D12450K/9F5020
I.
AUC
P=0.06
GTT
J.
IPITTT
% glucose drop
D12450K/9F5020
K.
AAC
ITT
L.
Male
Female
Water
Fructose
p-AKT
T-AKT
p-ERK
T-ERK
VIN
INSULIN
M.
p/t AKT
p/t ERK
Protein, fold change

Figure 6


A.
Male + W
Male + F
Female + W
Female + F
Liver weight (g)
D12450K/9F5020
*
****
P=0.08
B.
Liver to body (%)
ns
**
C.
Male
Female
Water
Fructose
Liver ORO
D12450K/9F5020
D.
Liver TG (µg/mg)
***
****
E.
Liver Cholesterol (µg/mg)
P=0.06
F.
KHK-C
KHK-A
VIN
G.
ACLY
ACC1
FASN
SCD1
H.
Protein, fold change (F/W)
Male - Fructose/ Water
Female - Fructose / Water
KHK-C
KHK-A
ACLY
ACC1
FASN
SCD1
I.
CPT1a
ACOX1
ACOT1
J.
Protein, fold change

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

Supplementary tables

Table 1. Mouse primers used for real-time quantitative PCR.

Gene	Sequence (5' – 3')	Forward/Reverse
Acly	CAGCCAAGGCAATTTCAGAGC	Forward
	CTCGACGTTTGATTAACTGGTCT	Reverse
Acc1	ACAGTGGAGCTAGAATTGGAC	Forward
	ACTTCCCGACCAAGGACTTTG	Reverse
Fasn	GGAGGTGGTGATAGCCGGTAT	Forward
	TGGGTAATCCATAGAGCCCAG	Reverse
Scd1	CAGCCGAGCCTTGTAAGTTC	Forward
	GCTCTACACCTGCCTCTTCG	Reverse
Khk-c	AACTCCTGCACTGTCCTTTCCTT	Forward
	CCACCAGGAAGTCGGCAA	Reverse
Cpt1a	AGTGGCCTCACAGACTCCAG	Forward
	GCCCATGTTGTACAGCTTCC	Reverse
Khk-a	TTGCCGATTTTGTCCTGGAT	Forward
	CCTCGGTCTGAAGGACCACAT	Reverse
18S	GTAACCCGTTGAACCCCATT	Forward
	CCATCCAATCGGTAGTAGCG	Reverse
Tbp	TGACTGCAGCAAATCGCTTGG	Forward
	ACCCTTCACCAATGACTCCTATG	Reverse
ACOX1	ACTCGCAGCCAGCGTTATG	Forward
	AGGGTCAGCGATGCCAAAC	Reverse
ACOT1	AGAGGAAGAGTTGGGCAGAG	Forward

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

	TTCGTCCCAGCAGCAGCG	Reverse

Table 2. Antibodies used for Western Blot.

Name	Citation	Supplier	Cat no.
Rabbit polyclonal anti-ACLY	RRID: AB_2223744	Cell Signaling Technology	4332
Rabbit monoclonal anti-ACC	RRID: AB_2219397	Cell Signaling Technology	3676
Rabbit monoclonal anti-FASN	RRID: AB_2100796	Cell Signaling Technology	3180
Rabbit polyclonal anti-SCD1	RRID: AB_823634	Cell Signaling Technology	2438
Rabbit monoclonal anti-KHK-A		Signal way Antibody	21708
Rabbit monoclonal anti-KHK-C		Signal way Antibody	21709
Mouse monoclonal anti-CPT1a	RRID: AB_11141632	Abcam	ab128568
Mouse monoclonal anti-Vinculin	RRID: AB_2272814	Novus Biologicals	NB600-1293
Rabbit polyclonal anti-OCTN2	RRID: AB_2191406	Proteintech	16331-1-AP
Rabbit polyclonal anti-CACT	RRID: AB_10642001	Proteintech	19363-1-AP

bioRxiv preprint doi: https://doi.org/10.1101/2024.11.26.625441; this version posted November 26, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

Rabbit polyclonal anti-CPT2	RRID: AB_2084849	Abcam	ab71435
Mouse monoclonal anti-ACADVL	RRID: AB_10609094	Santa-Cruz	sc-271225
Rabbit polyclonal anti-ACADL	RRID: AB_1859818	Abcam	ab82853
Mouse monoclonal anti-Actin	RRID: AB_476744	Sigma	A5441
Rabbit monoclonal anti-pAkt	RRID: AB_2315049	Cell Signaling Technology	4060
Rabbit monoclonal anti-total Akt	RRID: AB_915783	Cell Signaling Technology	4691
Rabbit monoclonal anti-pERK	RRID: AB_2315112	Cell Signaling Technology	4370
Rabbit monoclonal anti-total ERK	RRID: AB_330744	Cell Signaling Technology	9102
Mouse monoclonal anti-HADHA	RRID: AB_ 10862577	Abcam	ab110302
Mouse monoclonal anti-ACOX1	RRID: AB_ 3075447	Santa-Cruz	sc-517306
Mouse monoclonal anti-ACOT1	RRID: AB_10918465	Santa-Cruz	sc-373917